

MADHURA PRATAP HANDE DESHMUKH

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SUMMARY

Embedded Systems Engineer experienced in developing production-grade firmware and embedded software for real-time safety-critical platforms. Skilled in bare-metal firmware, low-level driver development, digital signal processing, and hardware-software integration. Interested in embedded systems, cyber-physical systems, and intelligent sensing technologies.

EXPERIENCE

Noccarc Robotics Private Limited Firmware Engineer

Bhosari, Pune, Maharashtra
June 2024 – present

- Built production-grade firmware and embedded software for Patient Monitor NX5, Humidifier RH07 and Ventilator V730i, on STM32 ARM Cortex-M microcontrollers, contributing to commercially deployed real-time embedded systems.
- Implemented low-level drivers and hardware abstraction layers for UART, SPI, ADC, DMA, GPIO, Timers, and external interrupts, enabling reliable interfacing with analog front ends (AFEs), sensors, displays, and peripheral devices.
- Engineered DMA-driven multi-channel data acquisition and DSP pipelines for physiological signal processing, achieving 98% measurement accuracy while reducing processing overhead by 20% across embedded monitoring applications.
- Implemented multithreaded embedded software on MPU-based systems using queues, semaphores, mutexes, software timers, and inter-process communication to support concurrent data acquisition, processing, and application workflows.
- Architected modular bare-metal firmware (20K+ LOC) using interrupt-driven execution, state-machine based control, watchdog recovery, fault handling, and reusable software components for scalable embedded systems.
- Delivered closed-loop PID control firmware for the RH07 Humidifier, completing the full development lifecycle from implementation and tuning to production deployment within a 75-day execution cycle.
- Resolved power management, battery charging, and hardware-software integration issues across embedded platforms, improving system reliability, reducing downtime by 30% and enabling stable operation under production conditions.
- Integrated firmware and embedded software with custom hardware in collaboration with hardware, validation, and clinical teams, improving verification efficiency, measurement consistency above 90% and overall product readiness.

Noccarc Robotics Private Limited Embedded Software Intern

Bhosari, Pune, Maharashtra
October 2023 – May 2024

- Implemented embedded software modules for Patient Monitor NX5, incorporating DSP-based digital filtering algorithms to improve signal-to-noise ratio by 35% and enhance physiological waveform quality.
- Designed automated PCB validation and production test jigs, reducing manual testing effort by 84% while improving manufacturing throughput test repeatability and production quality assurance.
- Debugged hardware-software integration issues using oscilloscopes, logic analyzers, JTAG, and protocol-level diagnostics to accelerate board bring-up, firmware validation and root-cause analysis during product development.

TECHNICAL SKILLS

Programming: C / C++, Embedded C, Python

Embedded Systems: Bare-metal, Register-Level Programming, Firmware Architecture, Hardware-Software Co-Design

RTOS & Operating Systems: FreeRTOS, CMSIS-RTOS, Linux (Ubuntu), Windows

Microcontrollers: STM32 (32-bit ARM Cortex-M0, M3, M4), MSP430 (16-bit RISC), ATmega (8-bit RISC)

Communication & Interfaces: UART, SPI, I2C, CAN, RS485, One-wire, ADC, DAC, DMA, PWM, Timers, GPIO, Interrupts

Signal Processing & Control: Digital Signal Processing, PID Control, Multi Sensor Data Acquisition

Development Tools & Debugging Tools: Keil uVision, Eclipse, VS Code, Git, Oscilloscope, Logic Analyzer, JTAG, ST-Link

Languages: English (professional), Hindi (fluent), Marathi (native), Japanese (JLPT N4)

EDUCATION

Government College of Engineering and Research, Savitribai Phule Pune University

Pune

Bachelor of Engineering in Instrumentation and Control

August 2019 - June 2023

- CGPA: 8.74/10

AWARDS & HONORS

- Savitribai Phule Pune University Rank 10, Instrumentation & Control Branch (2023)
- 2nd Place, COEP Tech I2I Innovation Challenge (2022) for Robotic Hand Prosthesis
- ABU Robocon: AIR 6 (2021), AIR 14 (2020)